

Abhishek Kumar Singh

Senior Data Scientist | Corporate Trainer

LinkedIn- <https://www.linkedin.com/in/abhishek-kumar-singh-8a6326148/>

Email- abhisheksingh007226@gmail.com

Phone- +91-8318238637

Job Summary:

A dynamic Data Scientist and Corporate Data Science Trainer with around 7 years of experience in training professionals and students across global clients. Demonstrated expertise in delivering comprehensive training sessions in Data Science, AI, Machine Learning, Python, Deep Learning, Data Analytics, and various industry tools and platforms. Proven experience working with leading organizations like Deloitte, Capgemini, Shell, Federal insurance of corporation, GSK, Truist, MetLife and Knowledge Academy, along with significant contributions to startups and universities. Known for providing hands-on, practical training that bridges the gap between theoretical knowledge and real-world application.

Technical Skills:

- **Programming Languages:** Python, R, SQL, PostgreSQL
 - **Data Science & Machine Learning:** Scikit-Learn, TensorFlow, Keras, NLP, Deep Learning, Data Visualization, MLOps Tools.
 - **Data Analytics & Visualization:** Tableau, Power BI, Statistical Modelling (AB testing, KS test and many more)
 - **Big Data Tools:** Hadoop, Apache Spark, PySpark, Kafka, Apache Nifi
 - **Cloud Platforms:** AWS, Azure, Google Cloud Platform (GCP)
 - **Version Control:** Git, GitHub
 - **Database Management:** SQL, NoSQL, RDBMS, Vector Database, Knowledge Graph
 - **Tools & IDEs:** Jupyter Notebook, VS Code, PyCharm
 - **Others:** Machine Learning Algorithms, Data Preprocessing, Data Cleaning, Model Deployment, Data Pipelines, Generative AI (RAG, Fine tuning), Agentic AI (CrewAI, LangGraph, Agno), Communication protocols (MCP, A2A)
-

Education:

- **B.Tech in Computer Science** - Lovely Professional University, GPA: 6.15
 - **Class 12th** - Gopal Inter College, 83%
-

Clients and Organizations Trained For:

I have delivered over 45+ corporate training (around 3000+ professionals) in various Organization like @Deloitte, @GSK, @Bechtel, @Shell, @Capgemini, @Metlife, @Truist, @JPMorgan, @Salesforce, @TheKnowledgeAcademy, @Wells Fargo, @Elevance health, @Federal deposit of Insurance corporation, @Department of Education USA, @Consumer financial protection bureau, @Travel + Leisure, @vguard and many more.

Training Experience:

- Trained over **3,000+ professionals** across the UK, USA, Greece, Australia, and India in Data Science, Engineering, and Machine Learning.
 - Designed and delivered comprehensive training modules covering topics such as Python, Data Analysis, Deep Learning, Machine Learning, and Big Data tools.
 - Specialized in creating custom training materials, hands-on exercises, and projects to ensure an engaging learning experience.
 - Successfully conducted training programs both in-person and remotely, adapting to different learning styles and requirements.
-

Work Experience:

- **DataSpoof (2020-Present)**
 - Collaborated with companies such as Alagents (Netherlands), L'Oréal (Australia), and multiple other international clients.
 - Managed software automation projects for appraisal reports and annual report extraction for Fortune 500 companies.
 - Delivered successful project management for a UK-based healthcare startup focused on facial disease identification.
 - Automated data pipelines and ETL processes for streamlined data management and quality.
 - Built an ETL Pipelines for 1TB of Twitter dataset using Spark and MapReduce. Engineered features leveraging tweet contents and user interaction to provide suitable recommendations.
 - Develop, train, and deploy machine learning models within Databricks to achieve the desired outcomes.

- Utilize Databricks MLFlow for model tracking, registration, and management.
 - Optimize machine learning workflows and troubleshoot any issues that arise during model development and deployment.
-
- **Freelance Machine learning engineer (2019-2020)**
 - Developed and deployed a GAN-based model for an Image-to-Image Translation project.
 - Built and deployed an Automated License Plate Recognition (ALPR) system capable of operating accurately across diverse environments, including day, night, cloudy, and low-visibility conditions.
-

Certifications:

- **Certified Data Scientist from Coursera**
-

Projects

Agentic AI & Generative AI Projects

End-to-End Financial Intelligence Platform using RAG

DataSpooof | April 2025 – November 2025

- Led the development of an **AI-powered financial intelligence platform** to automate analysis of SEC filings and multiple financial data sources, reducing manual research time for analysts by enabling natural language access to complex financial data.
- Designed and implemented a **scalable ETL pipeline** to ingest and process data from **20+ structured and unstructured financial sources**, transforming large-scale datasets using **PySpark** and storing curated financial data in **AWS RDS**.
- Built an **embedding and semantic retrieval system** using **Anthropic Claude and vector databases**, enabling analysts and financial teams to query regulatory filings and financial disclosures using natural language.
- Implemented **advanced Retrieval Augmented Generation architectures (Branched RAG and Corrective RAG)** to improve document retrieval precision and reduce hallucinations in financial insights.
- Collaborated with product and analytics teams to **optimize retrieval accuracy and response quality through A/B testing**, improving the reliability of AI-generated financial insights.
- Worked with on **AWS cloud infrastructure**, designing a scalable architecture using **API Gateway, ECS, EC2, CloudFront, and Route53**, enabling secure and reliable access to the

AI platform.

Tech Stack: Glue, RDS, API Gateway, Route 53, CloudFront, CloudWatch, ECS, EC2, Pinecone, Anthropic, LLMOPS, NLP, AB Testing, PySpark

Data Science Projects

Large-Scale ML Pipeline for Ad Click Prediction & Optimization

DataSpooof | Jan 2025 – Mar 2025

- Led the development of a **large-scale machine learning platform for ad click-through rate (CTR) prediction**, enabling optimized ad placement and improving monetization across **Samsung, Xiaomi, and Truecaller ecosystems**.
- Designed and implemented a **real-time data ingestion and feature pipeline** using **Kafka streaming**, processing high-volume user interaction events from millions of mobile users.
- Developed **high-performance CTR prediction models using LightGBM**, enabling low-latency inference for real-time ad targeting and recommendation.
- Deployed scalable ML models on **AWS SageMaker**, optimizing infrastructure utilization and supporting **billions of prediction requests at production scale**.
- Partnered with product and advertising teams to implement **A/B testing frameworks** for evaluating ad placement strategies, enabling data-driven optimization of **CTR and advertising revenue**.
- Built a **robust MLOps pipeline** using **MLflow, Prometheus, and Grafana** for automated model deployment, performance monitoring, drift detection, and system health tracking.
- Delivered a **production-grade ML system generating 1B+ predictions daily**, impacting **10M+ users** and supporting real-time decision-making for ad delivery and optimization.

Tech Stack: Kafka, LightGBM, AWS SageMaker, Prometheus, Grafana, Python, MLOps, NLP, AB Testing, PySpark

ETL Pipeline on AWS for Near-Earth Object (Asteroid) Monitoring

DataSpooof | Jan 2023

- Led the development of a **cloud-based data intelligence pipeline** to collect and analyze **Near-Earth Object (asteroid) data from NASA APIs**, enabling continuous monitoring of asteroid characteristics such as velocity, size, and proximity to Earth.
- Designed and implemented a **scalable AWS ETL architecture** to ingest high-frequency astronomical data using **AWS Kinesis Firehose**, process transformations via **AWS Glue**, and store curated datasets for analytics in **AWS Athena**.
- Engineered optimized **analytical data models** to support both **real-time monitoring and historical analysis** of asteroid trajectories and risk indicators.
- Built **interactive Grafana dashboards** enabling researchers and analysts to visualize asteroid movement patterns, velocity distributions, and proximity metrics.

- Enabled **data-driven scientific insights and monitoring capabilities**, supporting exploratory analysis and visualization for space research and anomaly detection use cases.

Tech Stack: AWS Firehose, AWS Glue, Athena, Grafana, Python, PySpark

Advanced Brain MRI Segmentation for Diagnostic Accuracy

DataSpoof | Aug 2021 – Sep 2021

- Collaborated with a cross-functional team of data scientists and engineers to develop an AI-driven medical imaging solution for accurate brain MRI segmentation, supporting clinicians in identifying critical anatomical structures and abnormalities.
- Designed and implemented deep learning segmentation models using U-Net and CNN architectures, combining them with traditional image processing techniques to improve boundary detection and reduce noise in MRI scans.
- Worked with the team to address key challenges in medical imaging such as boundary ambiguity, image noise, and segmentation accuracy, improving the reliability of automated diagnostic workflows.
- Conducted model validation and experimentation using A/B testing and causal inference approaches, ensuring robust evaluation of segmentation performance.
- Contributed to the development of clinical decision support capabilities, enabling more precise segmentation outputs for downstream diagnostic analysis and research.

Tech Stack: Deep Learning, UNet, CNNs, Medical Imaging, AB Testing, Casual Inference

Facial Disease Identification Using Deep Learning

DataSpooF | Aug 2021

- Collaborated with a **cross-functional team of data scientists and computer vision engineers** to develop an AI-based facial disease detection system aimed at supporting **early medical screening and diagnostic assistance**.
- Designed and trained **deep learning computer vision models (CNNs)** combined with traditional ML feature extraction techniques to detect facial patterns associated with multiple medical conditions.
- Enabled automated identification of **single and multi-disease cases**, including **beta-thalassemia, hyperthyroidism, Down syndrome, and leprosy**, using facial imagery.
- Optimized model performance through **feature engineering and CNN architecture tuning**, achieving **>97% classification accuracy** across disease categories.
- Evaluated model robustness using **A/B testing and causal inference methods**, ensuring reliable performance across diverse image samples and real-world scenarios.
- Contributed to the development of a **screening support tool for healthcare research and early disease detection**, enabling faster analysis of facial indicators linked to genetic and metabolic disorders.

Tech Stack: TensorFlow, Keras, Computer Vision, CNNs, AB Testing, Casual Inference

Achievements:

- Recognized for impactful training sessions, resulting in student placements and internships with top-tier organizations like **Deloitte**.
- Recognized from **VGuard** for training their 300 employees on GenerativeAI.
- **Recommendation** from Director of Truist Bank, Vice President of Watania Takaful and Senior Architect from Federal Deposit of Insurance Corporation.
- **Guest Speaker** at Jesus and Mary College, Kurukshetra University and Satyabhama University, sharing insights into data science, big data along with its applications.

Languages:

- **English:** Full Professional Proficiency
- **Hindi:** Native or Bilingual Proficiency

